

Oliver Dippel

Research Engineer · Reinforcement Learning & Foundation Models

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SUMMARY

PhD research engineer in reinforcement learning and foundation models, with peer-reviewed work at **AAMAS 2026** and **SGAI 2024/2023**. Specialises in **why training breaks**: reward hacking, posterior and entropy collapse, distribution shift. Diagnoses these through systematic ablation and statistical root-cause analysis rather than hyperparameter search, builds the distributed training stack the work runs on (JAX, PyTorch, Ray, SLURM), and ships the result into production. Deployed systems include a decision-support tool in sustained operational use and agentic systems in safety-critical and B2B settings.

EXPERIENCE

Technical Advisor, Agentic Systems · Curate IQ Jun 2026 – Present
Seattle, WA (remote) · independent, part-time

- Designed the intent-routing and RAG architecture for a B2B agentic platform, fusing catalog, safety and industry-knowledge sources so multi-skill hand-off resolves to the correct specialist agent.
- Built the retrieval pipeline on live product data, keeping every generated answer strictly grounded in the merchant catalog.

Research Engineer, RL & Foundation Models (PhD) · University of Liverpool Dec 2021 – Jun 2026
CDT in Distributed Algorithms · industry-funded

- Designed the **Heuristic Transformer**, a Bayesian RL architecture that improves zero-shot policy generalization over standard RL baselines on peer-reviewed benchmarks (AAMAS 2026).
- Diagnosed and resolved recurring training-time pathologies — **reward hacking, posterior collapse, entropy collapse** — through systematic ablation studies and statistical root-cause analysis, not hyperparameter patching.
- Deployed a model fine-tuned on structured expert feedback as a real-time recommender; it remains **in sustained operational use** as a decision-support tool among domain specialists.
- Cut experiment turnaround across HPC job cycles by architecting fault-tolerant, reproducible distributed training and MLOps pipelines in JAX and PyTorch on Ray and SLURM.

Research Engineer, LLM Agents & Agentic Systems · ZeroAI Sep 2023 – Dec 2024
Liverpool, UK · part-time, concurrent with PhD

- Built a drift-detection and regression-testing harness that surfaced distinct agent failure modes under distribution shift **before deployment**, via a systematic adversarial evaluation suite.
- Integrated diffusion models into a production perception pipeline, improving multi-step trajectory forecast stability in high-dimensional sensor state-spaces.
- Engineered tool-use and structured-output validation on AWS for a **safety-critical autonomous-driving** agent system, with CI/CD-integrated monitoring and observability.

Statistical Consulting & ML Teaching · statistiknachhilfe.ch Jan 2021 – Present
Freelance

- Designed and taught a curriculum in Bayesian and probabilistic ML for technical and non-technical audiences.

Data Science Intern, Applied ML · Novartis Pharma AG Jun – Dec 2020
Basel, Switzerland

- Validated ML pipelines against GxP data-quality standards in a regulated healthcare environment.

SELECTED PUBLICATIONS

Heuristic Transformer: a Bayesian RL architecture for zero-shot policy generalization. **AAMAS 2026**. With A. Lisitsa, B. Peng.
Contextual Transformers for Goal-Oriented Reinforcement Learning. **SGAI 2024**. With A. Lisitsa, B. Peng.
Deep Reinforcement Learning for Continuous Control of Material Thickness. **SGAI 2023**. With A. Lisitsa, B. Peng.

SKILLS

Research: Reinforcement learning (policy gradients, Bayesian RL, exploration, credit assignment), RLHF and reward design, LLM fine-tuning (SFT, LoRA/PEFT), diffusion models, distribution-shift and drift diagnosis, applied statistics
Agentic systems: Tool-use agents, multi-step reasoning, retrieval-augmented generation, structured output, multi-agent orchestration, evaluation and regression harnesses
Engineering: Python (advanced), JAX (Haiku/Flax), PyTorch, Ray, SLURM/HPC, AWS, Kubernetes, Terraform, Docker, Weights & Biases, Git, CI/CD
Languages: German (native) · English (fluent) · French (intermediate)

EDUCATION

Ph.D. Electrical & Control Engineering — University of Liverpool, CDT in Distributed Algorithms 2021 – 2026
M.Sc. Applied Statistics — University of Göttingen 2017 – 2020
B.A. Economics — University of Göttingen 2014 – 2017